TeachEngineering

Modeling the Amygdala

Background

The following lesson is based on the work of Li, Nair, and Quirk in 2009.
https://doi.org/10.1152/jn.90765.2008

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ARTICLES

A Biologically Realistic Network Model of Acquisition and Extinction of Conditioned Fear Associations in Lateral Amygdala Neurons

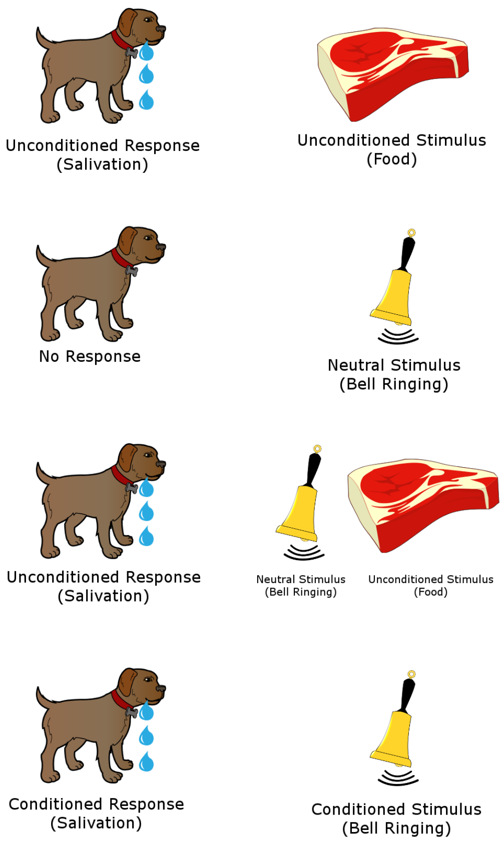
Guoshi Li, Satish S. Nair, and Gregory J. Quirk

01 MAR 2009 // https://doi.org/10.1152/jn.90765.2008

Review: Classical Conditioning

Think back to lesson 1 and learning/conditioning. We will be applying this to an experiment where rodents learn to associate a sound with fear.

- An unconditioned (innate) response
  - For Pavlov, dogs salivated in response to expecting food
  - For us, this is a fear response to expecting a shock
- An unconditioned (neutral) stimulus
  - For both our and Pavlov's experiments, this is a sound
- The process of conditioning, where the innate response is associated with the neutral stimulus
- We will be using this process to explore the fear circuit in the amygdala



Ivan Pavlov's famous conditioning experiment

Measuring fear

Thing about how you would categorize fear in humans.

What characteristics could you universally use to determine the level of fear a human is experiencing?

Rodents as a Model System

The answer is we don’t know. Fear in humans is too complicated and the responses are too diverse to be able to accurately measure it.

This is why we use rodents to model fear!

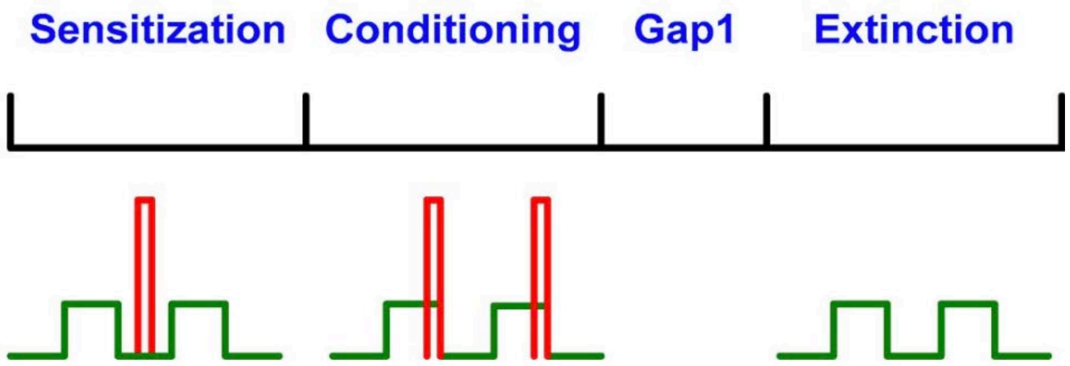
- They have very distinct and universal reactions to fear: they freeze
- Scientists have developed algorithms that can look at a video and score the fear response of the rodents

Experiment Phase 1 – Sensitization

- There are four phases: Sensitization (10 trials), Conditioning (10), Gap1 (none; sleep day 1 night), Extinction (30). During each phase, the rodent is subjected to tone-shock ‘trials’ (each 4 sec: 0.5 sec tone/shock+ 3.5 sec gap).
- In the sensitization phase, 10 tones and 10 shocks are delivered to the model ‘unpaired’, i.e., tone occurs in between, rather than during shock (see below). This is to get a control/baseline.
- Recent studies only deliver tones in this phase, i.e., no unpaired shock, called habituation. That is, to ensure that the rodents do not react to the sound, i.e., they get habituated to sound.
- The figure below shows the ordering of the tone and shock at each phase of the experiment

RED = shock

GREEN = tone



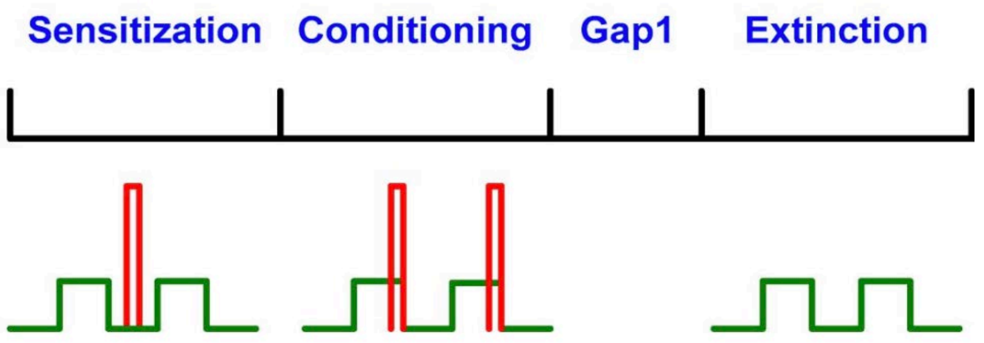
A graph showing the order of tone and shock at each phase of the experiment

Experiment Phase 2 – Conditioning

- You may have noted from the video that rodents respond to fear by freezing. The amount of duration that they freeze is measure automatically from the video by a program.
- In the next phase, the model is subjected to 10 conditioning trials of 4 seconds each which is where the association between auditory and pain stimuli is learned: 500 ms tone + 3.5 second gap. If shock is present, it overlapped with the last 100 ms of tone, i.e., from 400 – 500 ms of tone.
- Think about **Hebb's Law**: "cells that fire together wire together". By having the shock and tone overlapping for a period of time, we allow the neurons to fire at the same time. How is this helping learning – recall Colab on pairing and Ca-learning rule!

RED = shock

GREEN = tone



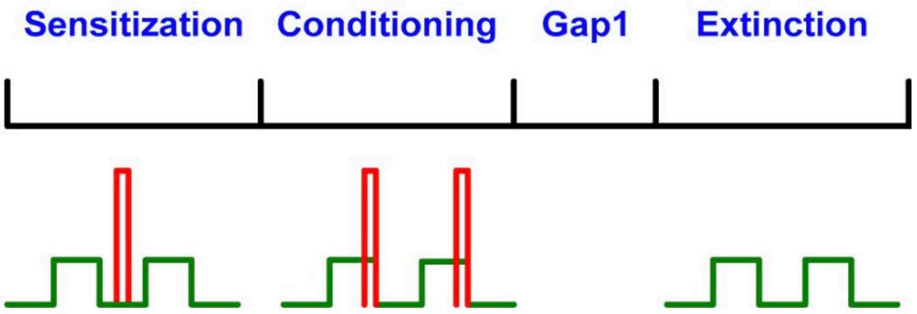
A graph showing the order of tone and shock at each phase of the experiment

Experiment Phase 3 – Extinction

- In the extinction phase, the rodent (or human) learns that tone may not always be paired with shock.
- In extinction, 30 tones are played with no shock
- Once this data are collected, the baseline firing rate from the sensitization phase can be subtracted out from the firing rate after conditioning to determine the ‘learning’, i.e., effect of conditioning. Subtracting the control from the experimental data is common practice in research to determine the effect of the experiment, as illustrated in later slides.

RED = shock

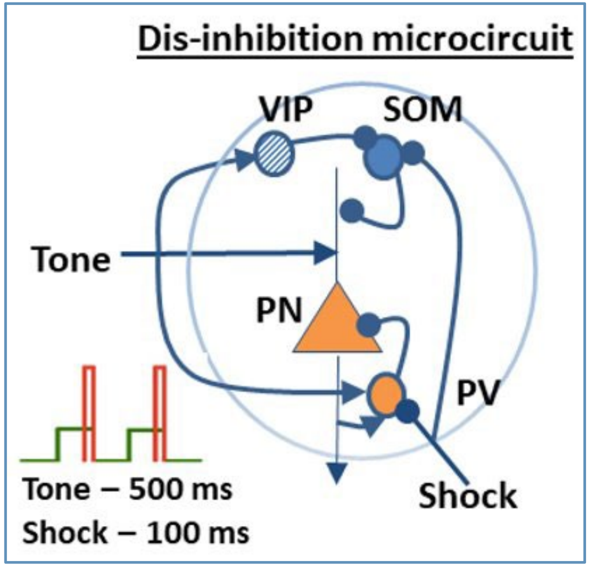
GREEN = tone



A graph showing the order of tone and shock at each phase of the experiment

Review of Neuron Types

- Different types of cells play unique roles in neural circuits by exciting or inhibiting other cells.
- Pyramidal Neurons **(PN)**: Main excitatory neurons, which connect across regions and so main transmitters of information in the brain. So, called principal neurons.
- Some PNs are adapting type and fire only a few spikes at the beginning of the stimulus, while others fire throughout the stimulus and are called tonic firing neurons.
- Parvalbumin **(PV)** neurons : Inhibitory interneurons to regulate the excitement of PNs and other cells. These connect only locally, typically in a 300 micron radius
- Somatostatin **(SOM)** neurons: Interneuron that inhibits PVs (disinhibition) and PNs. Again, only local connections.
- There’s an additional VIP cell, this is a type of interneuron that we will not be covering in this course.



- Interneurons (PV, SOM, VIP) serve to regulate the PN
- Shock inhibits PV & SOM (disinhibiting the PN)
- VIP gates the tone input to regulate the PN's excitation

Moving on to a more realistic amygdala network model

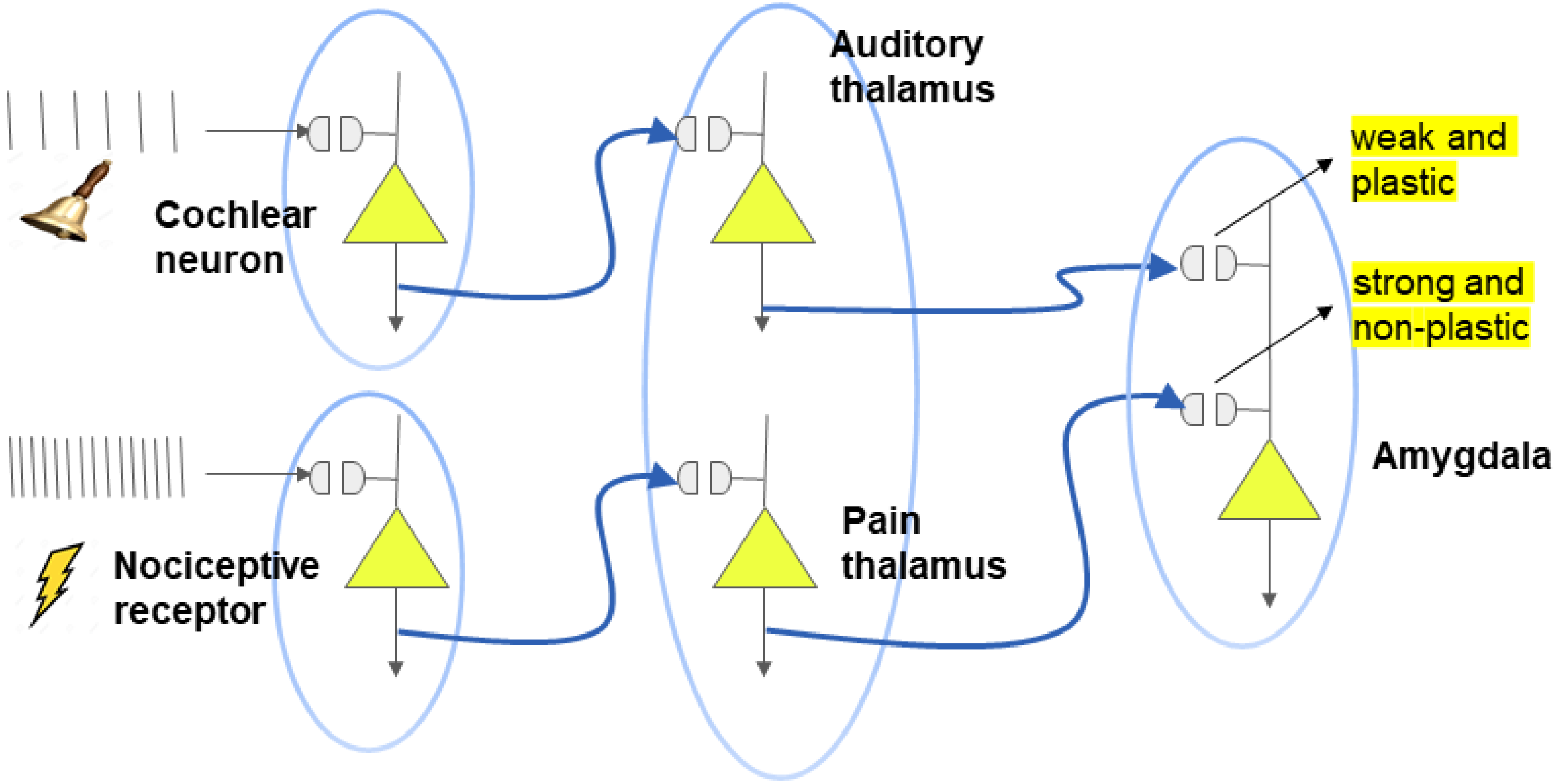
You sketched the simplified fear circuit (total of 5 neurons along the tone-shock neural pathways)

……….

Now we move on to a more realistic version of the fear circuit similar to those used by researchers engaged in reverse engineering brain circuits!

But before we start, we have an activity where you sketch your understanding of how neurons are 'wired' in our brains.

We move on to a larger and realistic model from your earlier sketch (below)

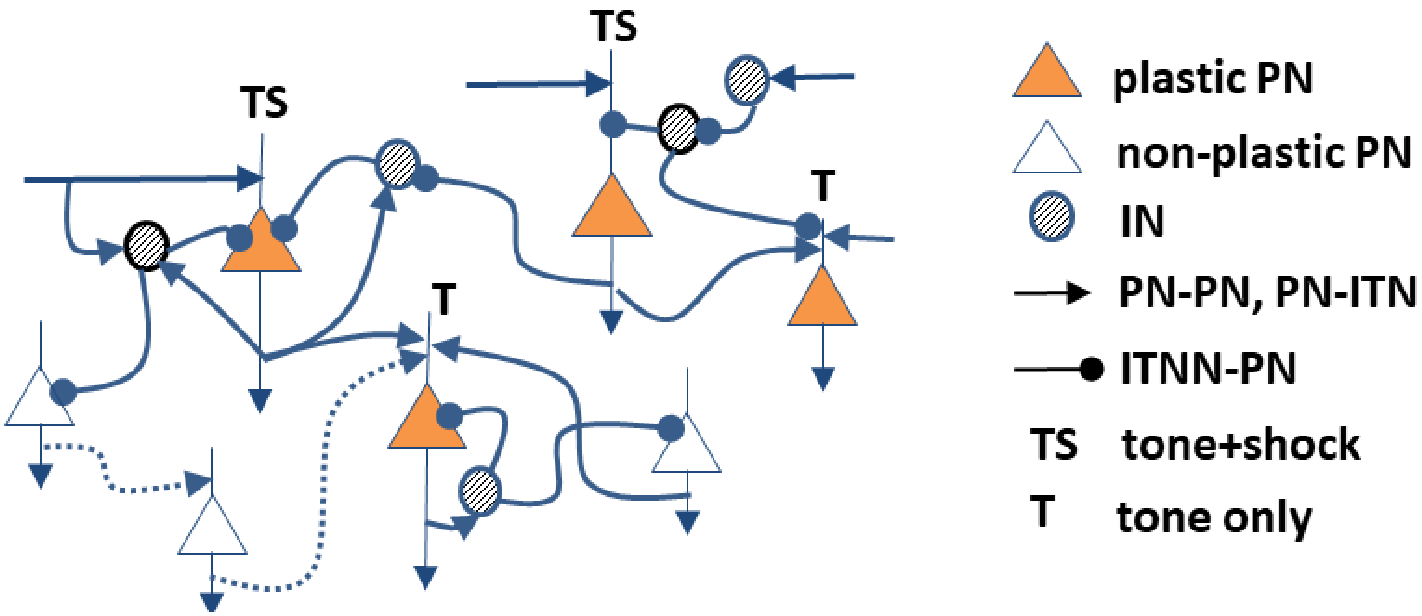


 Your student ID Number ____________

ACTIVITY 1a – you sketch a larger amygdala network

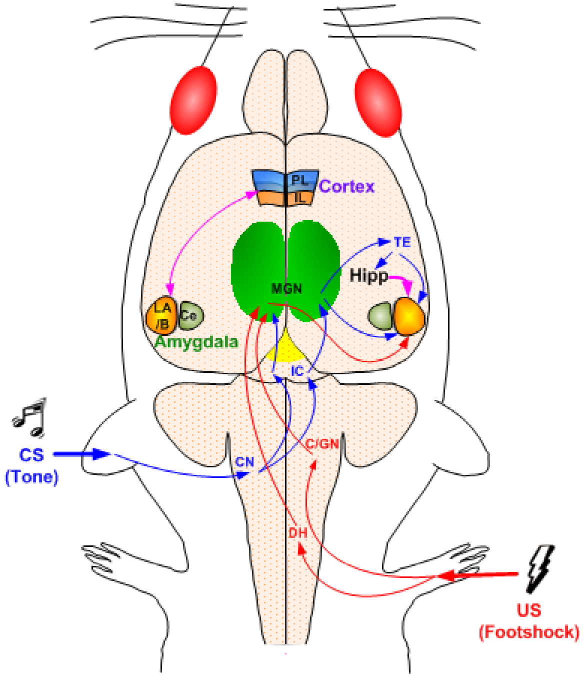
Exercise: Sketch 8 PNs and 2 ITNs (we will consider only one type, for simplicity) and connect them as connected in our brains. Use triangle symbol for PN, and circle for ITN. Use T for tone and S for shock and assume both make excitatory synapses. Use all the motifs in the previous lesson except the dis-inhibition one to not clutter the schematic. Excitatory synapses → and inhibitory as ---o

Some basics for your sketching

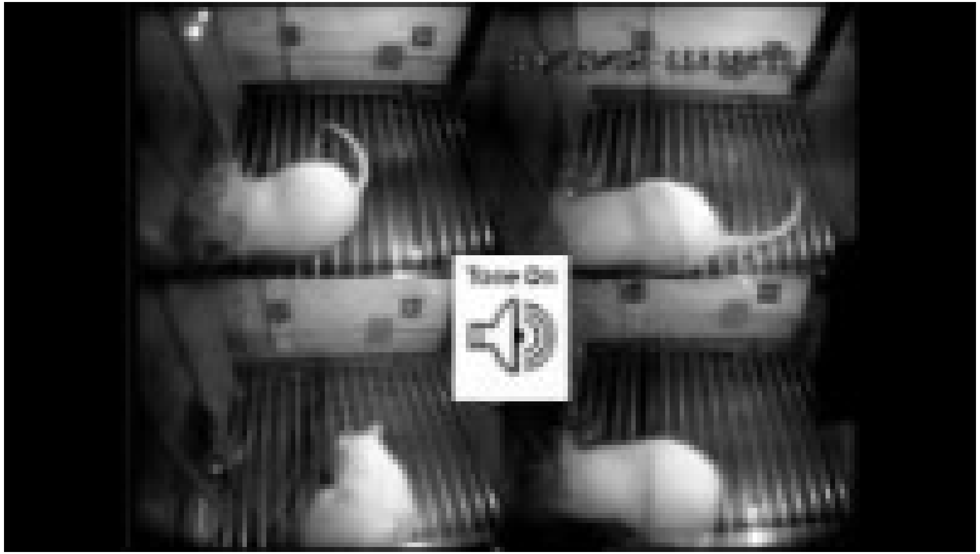


- You will sketch a 10-cell network model using the microcircuit motifs in the tutorials D1-D3 + then run a model to study how your amygdala learns fear
- Expands the simplified fear circuit you sketched earlier

Quick review – what are we modeling?



Pathways involved in auditory fear conditioning. Tone and shock inputs converge in the amygdala which learns to 'associate' them - "fear memory" stored in the amygdala

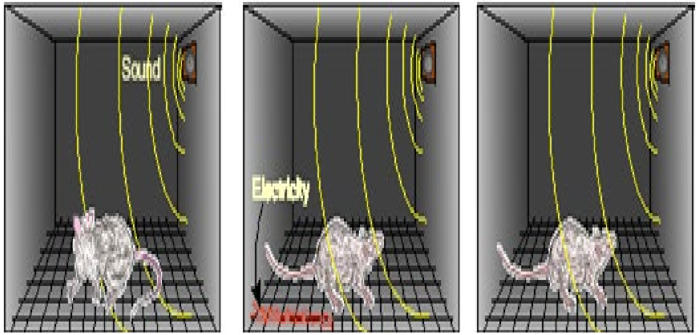


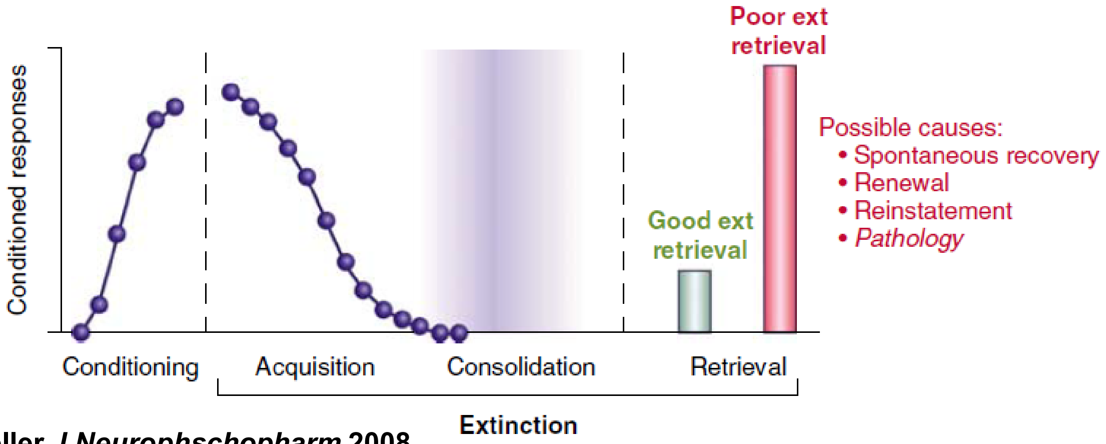
https://www.youtube.com/watch?v=mi2gqhHw1N0

Long term goal – identifying potential targets for pharmacological interventions. Relevant to anxiety disorders including post-traumatic stress

Fear conditioning & extinction

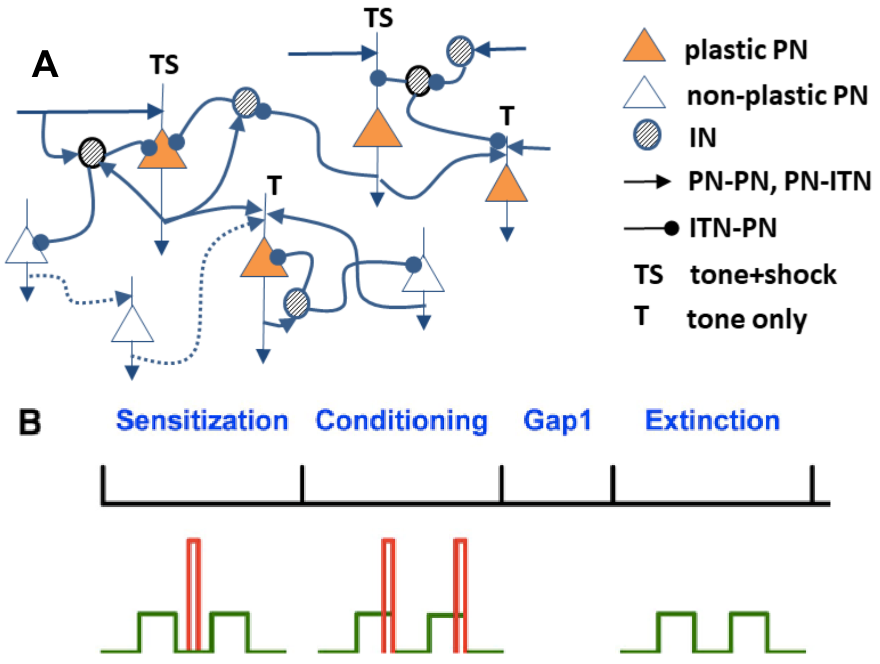
Rat conditioned to fear a tone





Quirk and Mueller *J Neurophschopharm* 2008

Some basics for your sketching….

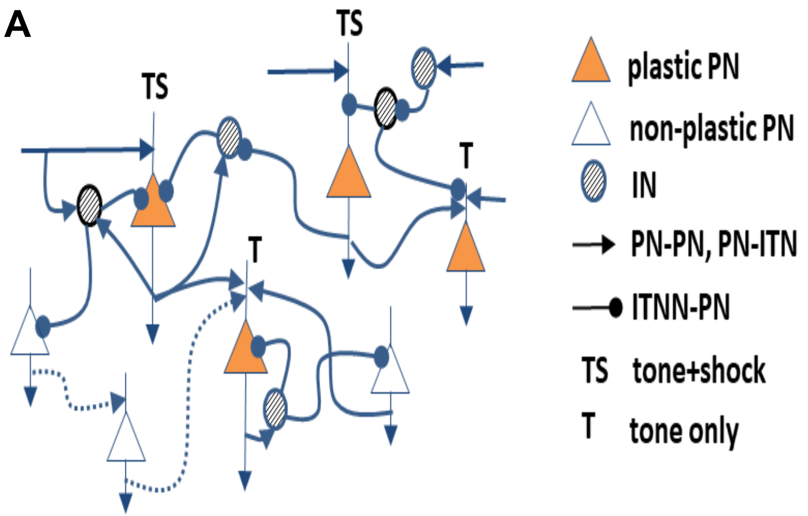


- Two types of neurons: excitatory (pyramidal, PN) and inhibitory interneurons (ITN)
- 8 PNs (~80%) and 2 ITNs (20%).
- The connectivity between pyramidal neurons is ~10%, both PN-ITN and ITN-PN connectivity are ~30%, and the same for ITN-PN connectivity. Also, ITN-ITN connectivity is ~30% in the amygdala.
- The model has excitatory AMPA/NMDA synapses and inhibitory GABA synapses
- Tone and shock are distributed to random 70% PNs and to all ITNs in this model.
- Training protocol is shown in part B. Tone is in green and shock is in red.

A. Architecture of the amygdala network model with cell types, connectivity and intrinsic and extrinsic inputs.
B: simulation schedule showing tone (green – tone delivered during the period) and shock (red – shock during the period) inputs during the phases of sensitization (no pairing), conditioning (both paired), a gap (with subject resting/sleeping), and then extinction (only tone, no shock) on the next day.

Some basics ….contd.

- **SINGLE CELL MODELS** – designed using electrophysiology data that fully characterizes the single cells. So, there are no free parameters to tune**.**
- **SYNAPTIC WEIGHTS** – accurate estimates of these are typically not available, and so these are adjusted by the user to provide the desired output, i.e., "tuned".
- **EXTRINSIC INPUTS** – information about extrinsic inputs has to be obtained from the neurobiological literature.
- **Present model** – Distribution of T, S inputs, and synaptic weights, are different among cells. Being a small model, we opted for an all-to-all connectivity

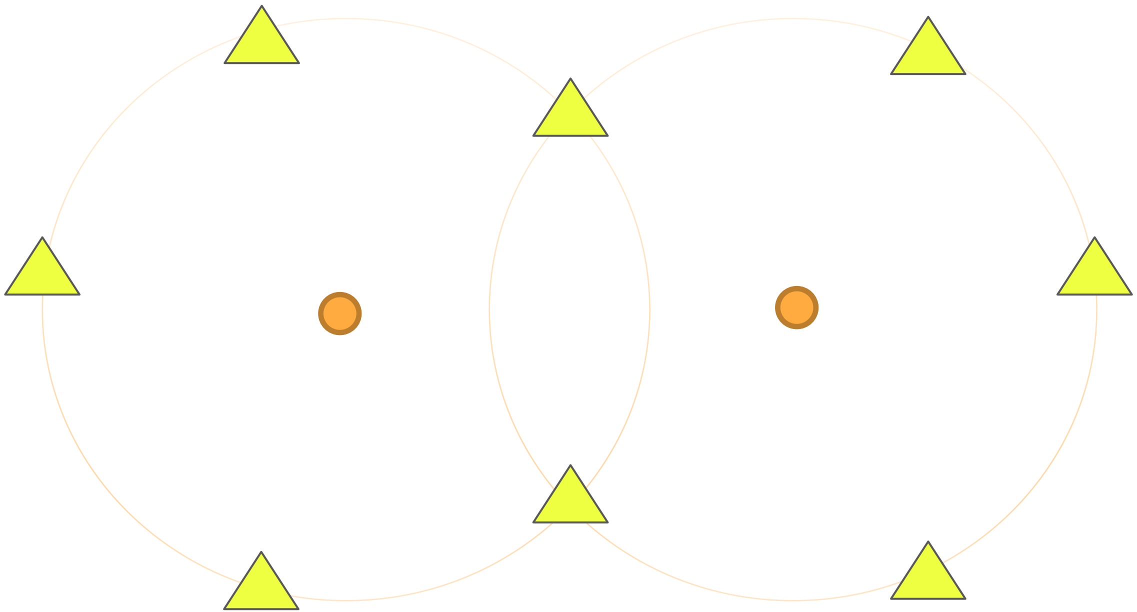


TO DO: After this review + explanation of the principles related to connectivity, you are to improve the sketch you made earlier. This will enhance your understanding of the 'rules' of connectivity, and provide you with circuit-level understanding of how a rodent's amygdala network might be learning fear

Your student ID Number _____________

ACTIVITY 1b – Repeat the sketch of the 10-cell amygdala network

Exercise: Sketch 8 PNs and 2 ITNs and connect them per rules given below. Use triangle symbol for PN and circle for ITN. Use T for tone and S for shock and assume both make excitatory synapses. Use all the motifs in the previous lesson except the dis-inhibition one to not clutter the schematic. Excitatory synapses → and inhibitory as ---o .



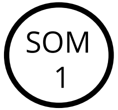
- Excitatory (pyramidal, PN) △ and inhibitory interneurons (ITN) ○
- 8 PNs (~80%) and 2 ITNs (20%)
- The connectivity between pyramidal neurons is ~10%, both PN-ITN and ITN-PN connectivity are ~30%, and the same for ITN-PN connectivity. Also, ITN-ITN connectivity is ~30% in the amygdala
- Tone and shock are distributed to random 70% PNs and to all ITNs in this model. To avoid clutter, write T or S or both next to the neuron to indicate this, i.e., do not draw lines.

How to model learning in amygdala using the model?

- Fear learning happens in the amygdala, and four major types of neurons are involved (<u>note: we did not consider the SOM cells earlier, to avoid complexity. Added here now</u>)
  - Adapting pyramidal (principal) cells 

  - Tonic firing pyramidal cells 

  - Parvalbumin fast-spiking inhibitory interneurons 

  - Somatostatin low-threshold spiking inhibitory interneurons 

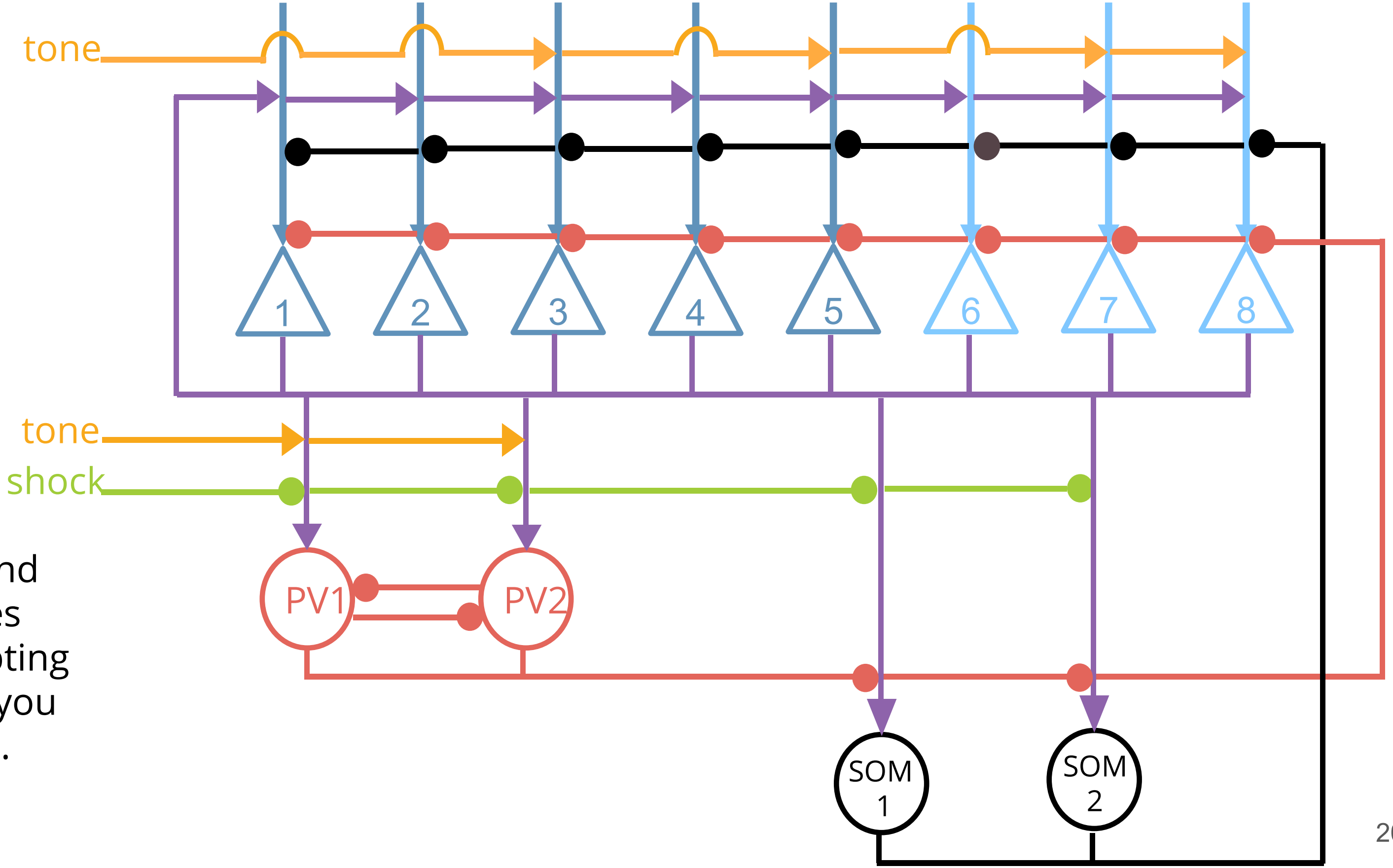
- There are two kinds of synapses 
  - Excitatory (glutamatergic) synapses 
  - Inhibitory (GABAergic) synapses
- In the next activity, you start with predicting how the various neurons in the next slide will fire in sensitization vs. conditioning, based on tone/shock inputs and cell type, based on the connectivity shown there?

Activity 2: Making Predictions

1. What cells fire during sensitization?

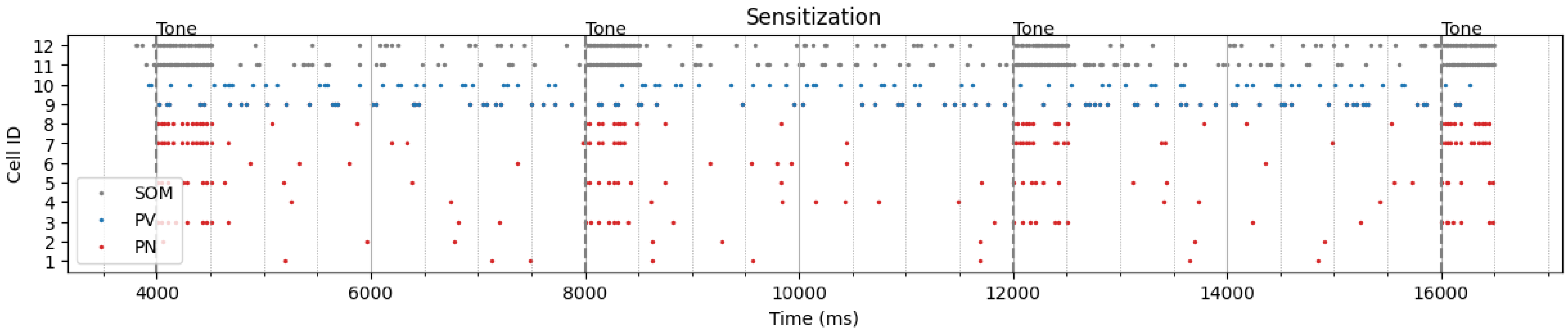
2. What cells (i.e., synapses to them) grow after conditioning?

Think about what combination of tone and shock each cell receives and the cell type (adapting vs tonic firing) to help you make your predictions.



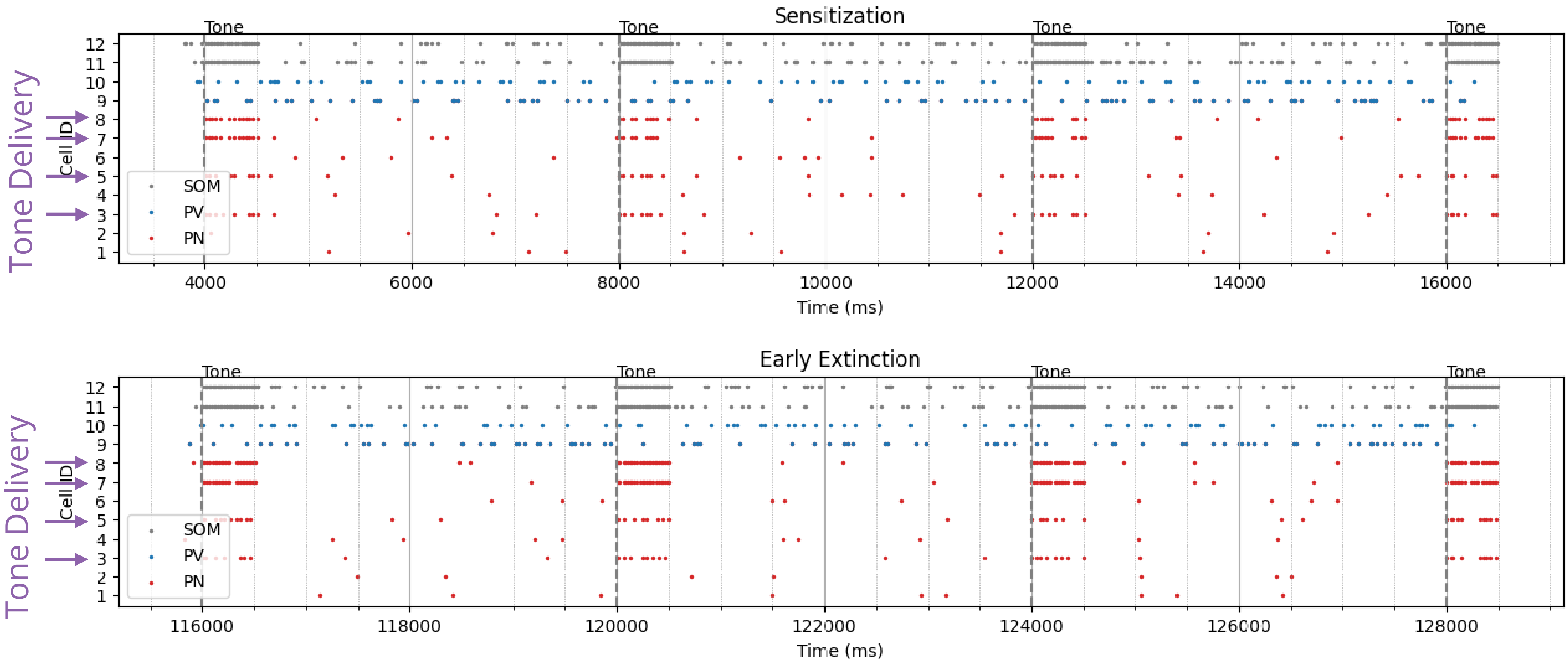
Gathering data: raster plot

- The spiking of each neuron is recorded in a **raster plot**. PNs are in red, PVs in blue and SOM gray
- A raster plot shows the spiking of all neurons over time. The x-axis is time (in ms), and the y-axis is the neuron number (total of 12 in this experiment; 8 PNs, 2 PVs and 2 SOMs)
- Shown below is a sample raster plot from the sensitization phase. Each dot represents when in time that particular neuron spiked during the trial. Three 4-sec trials are shown below: 4,000-8,000 ms, 8,000-12,000 and 12,000-16,000 ms.
- Note that some neurons fire more than the others. Can you guess why using the connectivity plot in the previous slide (#11)?.... will get clearer later.



Reading and interpreting raster plots

Three 4-sec trials are shown below: 116-120K ms, 120-124K ms, and 124-128K ms.



Activity 3 – Virtual Lab to simulate fear training

- Open this notebook (clicking on this should take you directly to the Colab version on your browser) that illustrates fear learning in a network model of the amygdala

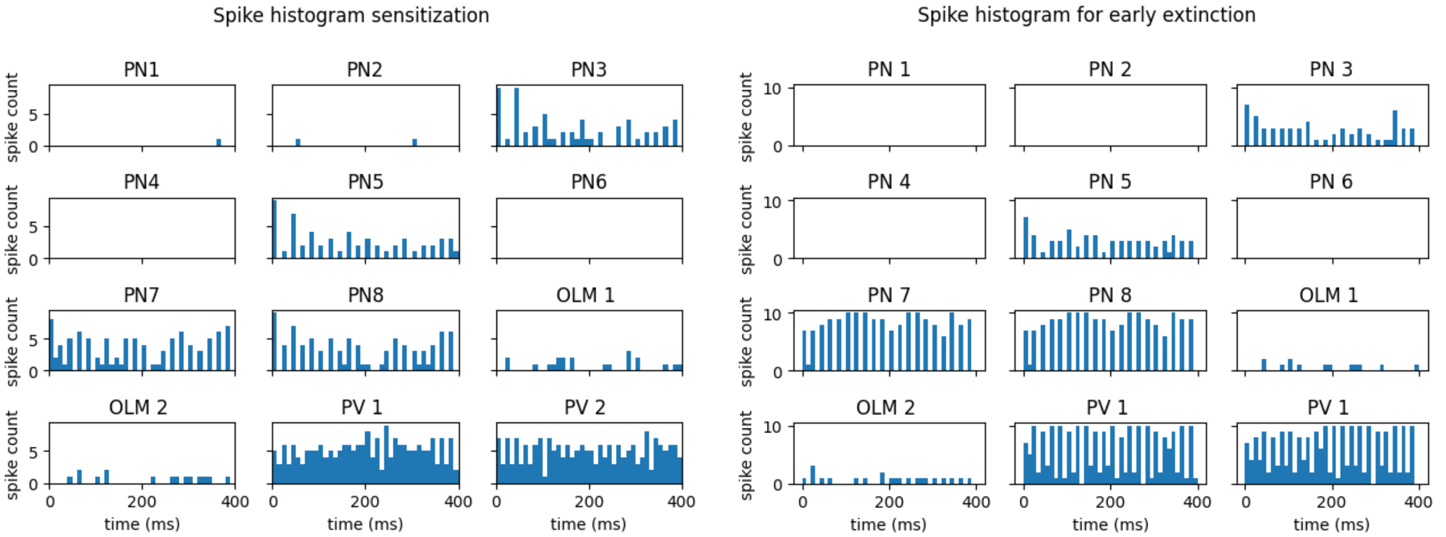
- Get familiar with the items on the sliders. The user can change tone weights to PNs, ITNs (PV and SOM), three connections between cell types, and three weights of the outside inputs

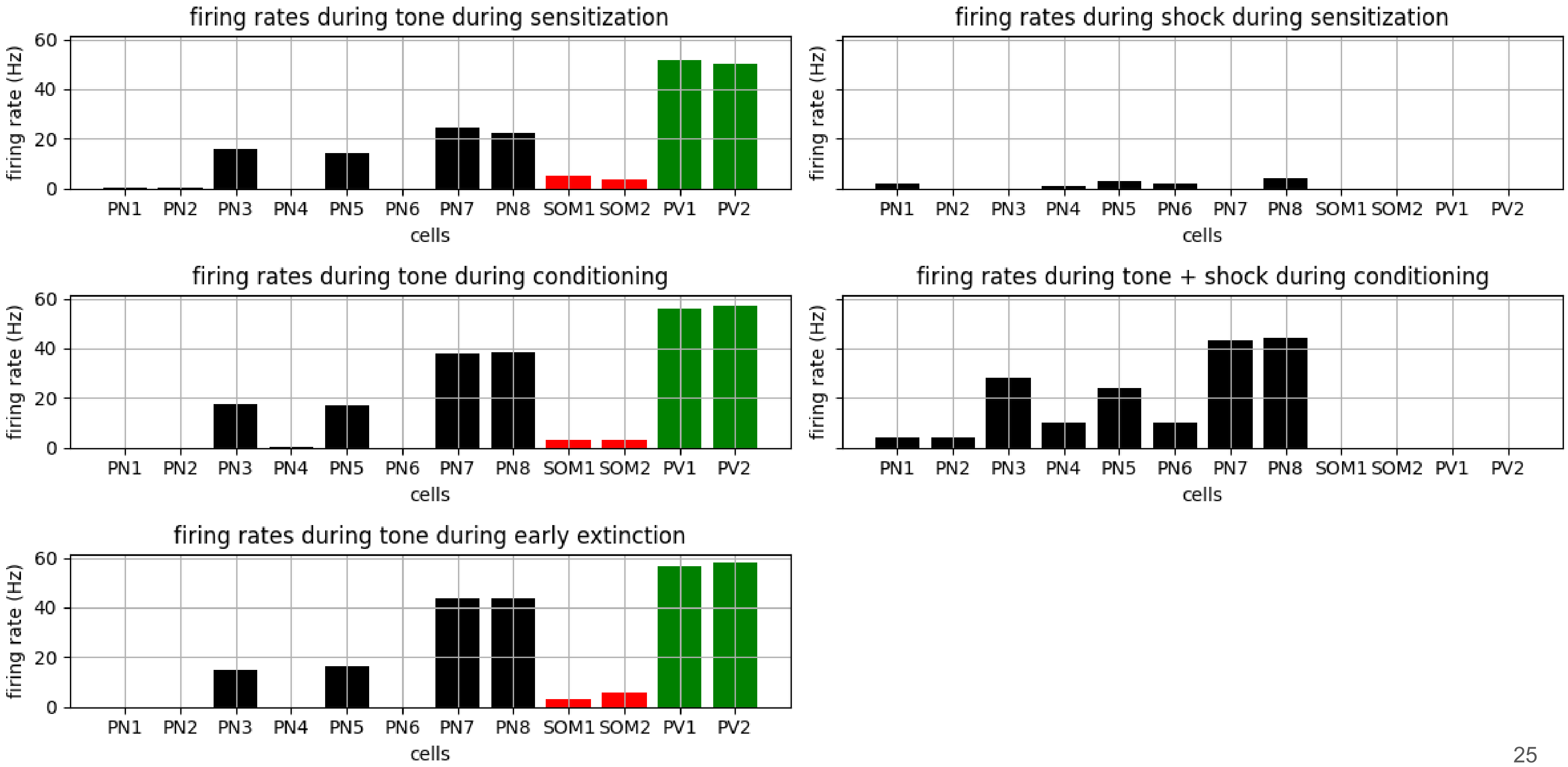
- Hypothesize how the firing rate of each neuron might change in network model of slide 20 with:
  - **Increased INT2PN**: greater connectivity between interneuron and pyramidal neuron
  - **Increased PN2PN**: greater connectivity between pyramidal neurons
  - **Decreased PN2PN**: less connectivity between pyramidal neurons
  - **Increased shock2INT**: greater synaptic weight of the shock input to interneurons
  - **Increased tone2PV**: greater synaptic weight of the tone input to pyramidal neurons

- Answer all questions above on a sheet of paper

How to interpret the data - baseline case of model

- Starting with a raster plot, we average the number of spikes of a given neuron over a 10 ms period (time bin of 10 ms) in each trial. This is how we go from raster plot → **bin plot**
- Can you guess why some neurons have higher spike counts per bin compared to others using the connectivity plot in a previous slide (#11)?




firing rates during tone during sensitization
firing rates during shock during sensitization
firing rates during tone during conditioning
firing rates during tone + shock during conditioning
firing rates during tone during early extinction
firing rate (Hz)
cells
0
20
40
60
PN1
PN2
PN3
PN4
PN5
PN6
PN7
PN8
SOM1
SOM2
PV1
PV2

Homework Questions

- Complete Activity 1 (slides 12 and18) - Sketches of connectivity for a 10-cell model of the amygdala

- Complete Activity 2 (slide 20) - Using the network model with connectivity plot shown on slide 13, make predictions on the specific input synapses that will grow or decay, causing the particular cell to increase/decrease firing. Provide the logic to support your prediction.
- List whether your prediction of the change in firing rate was consistent with the observed results from the model by using the pre-run data provided (when you click on the various buttons). Do this for each neuron type separately.
- How do you convert the bin plots of slide 24 to the firing rate plots on slide 25? Provide a few examples using the plots to consolidate your understanding.

- Complete Activity 3 (slide 23) – Given the model, you predicted which synapses may grow in the earlier activity. Now, you are to hypothesize the effect of changing connection strengths for various types of synapses on learning fear. Compare with model predictions to the extent possible.

You can check out the Li et al. (2009) paper on which this learning module is based.

Vocabulary

Sensitization: Phase 1 of our study where tones and shocks were randomly played; this acted as a baseline/control off of which we could determine the effect of conditioning

Conditioning: Phase 2 of our study where the shocks were played at the end of each tone; this is the period when learning occurs

Early Extinction: Phase 3 of our study where only tones are played; subtracting the sensitization results from this phase allows us to study neuron firing during conditioning

Hebb's Law: Cells that fire together, wire together

Pyramidal neurons (PN): Main excitatory neurons, transmit information in the circuit

- **Slow adapting**: Neurons that are not good at adapting to new sensory information; will keep firing throughout sensitization phase
- **Fast adapting**: Neurons that a good at adapting to new sensory information; will stop firing after a short period of time during sensitization phase

Vocabulary, cont.

Parvalbumin (PV): Inhibitory interneurons to regulate the excitement of PNs and other cells

Somatostatin (SOM): Interneuron that inhibits PVs (disinhibition) and PNs

Raster Plot: Plot displaying when each neuron in the model fires over a period of time; a plot of neuron ID vs time

Bin Plot: Plot created from a raster plot that shows a single neuron's spiking rate over time; a plot of the number of spikes vs time

Methods: Section describing the setup of the study, including how and what the researcher(s) will be measuring

Results: Section describing what the researcher(s) found after analyzing their data and how this relates back to their hypothesis

Discussion: Section where the researcher(s) interpret their data and pull together all the other sections of the paper; includes a bigger picture of study